

Geo-referenced small object detection in oblique aerial imagery

Paolo Moloney¹, Aidan Magee¹, Daire Walsh¹, Tim McCarthy¹, and John McDonald²

¹*National Centre for Geocomputation, Maynooth University, Co. Kildare Ireland*

²*Department of Computer Science, Maynooth University, Co. Kildare, Ireland*

Abstract

Unmanned Aerial Vehicles (UAVs) are increasingly deployed in emergency response operations, resulting in growing volume and complexity of aerial video which creates a critical situational awareness bottleneck for incident commanders. We present an end-to-end Geospatial-aware ML framework that fuses computer vision with terrain-aware geospatial processing to automatically detect, localise, and visualise objects of interest from drone-borne video streams. The system integrates a scalable ML inference backend, a terrain-aware geo-positioning pipeline, and a suite of operator-focused visualisation methods, transforming continuous aerial video into structured, geo-referenced intelligence that incident commanders can act upon without reviewing raw footage. Additionally, we carry out a preliminary performance evaluation stratified by sensor-to-target distance on a small held-out dataset, to characterise the operational envelope of the system in actionable terms.

Keywords: UAV, aerial object detection, geospatial AI, emergency response, situational awareness

1 Introduction

Time-constrained emergency response operations — wildfires, search-and-rescue, flood assessment — increasingly rely on drone-borne imagery to build situational awareness over terrain that is dynamic, potentially hazardous, and often unfamiliar to first responders. While UAVs provide continuous aerial surveillance at low cost, their intelligence value is largely unrealised in practice: operators must manually scan hours of video to locate personnel, vehicles, or structural hazards, creating a critical information bottleneck at exactly the moment when rapid decisions are most consequential.

Several compounding factors make this a hard technical problem: drone-generated data volumes exceed what operators can parse in real time, and existing computer vision systems typically lack the geospatial integration incident commanders need for map- and GIS-based workflows.

This paper presents an end-to-end framework addressing these challenges through three core pillars - Operations, ML & Geospatial Processing, and Visualisation. The primary contribution of the framework is the integration of established components into a deployable pipeline, together with an evaluation methodology framed in operationally meaningful terms.

2 Related Work

Oblique aerial imagery is preferred over nadir capture for Search and Rescue (SAR) as an oblique frame covers a substantially larger ground area, reducing survey passes required. However it presents distinct detection challenges: altitude leads to reduced pixel footprints for distant objects, and oblique viewing angles introduce extreme scale variation across and within frames, which are not present in ground-level or nadir imagery.

The VisDrone dataset [Zhu et al., 2021] has become a popular benchmark for this domain, with approximately two thirds of objects falling under the COCO small category (bounding boxes $< 32^2$ px).

[Akyon et al., 2022] proposes a generic framework for small object detection based on partitioning images into overlapping patches at both the fine-tuning and inference stages, increasing the relative pixel coverage of small objects within each patch. During Slicing Aided Fine-tuning (SF), training data is augmented with resized crops alongside the original images, preserving sensitivity across object scales. During Slicing Aided Hyper Inference (SAHI), the same slicing strategy is applied to query images, with slice predictions merged back into the original coordinate space via Non-Maximum Merging (NMM). They find that existing models can benefit from SAHI without additional fine-tuning, but gains can be maximised by combining with SF.

Converting a 2D detection in the image into a real-world ground location requires resolving where the camera’s line of sight to that pixel intersects the terrain. The dominant approach is to cast a ray from the camera’s optical centre using attitude and gimbal telemetry, then intersect with a Digital Elevation Model (DEM) to recover a ground coordinate. This is validated by [Paulin et al., 2024] in real world SAR scenarios, and by [Rossi et al., 2025] in thermal hotspot geolocation in wildfire contexts.

3 Methodology

The framework is structured around three pillars - Operations, ML and Geospatial Processing, and Visualisation. Figure 1 gives a high-level system overview.

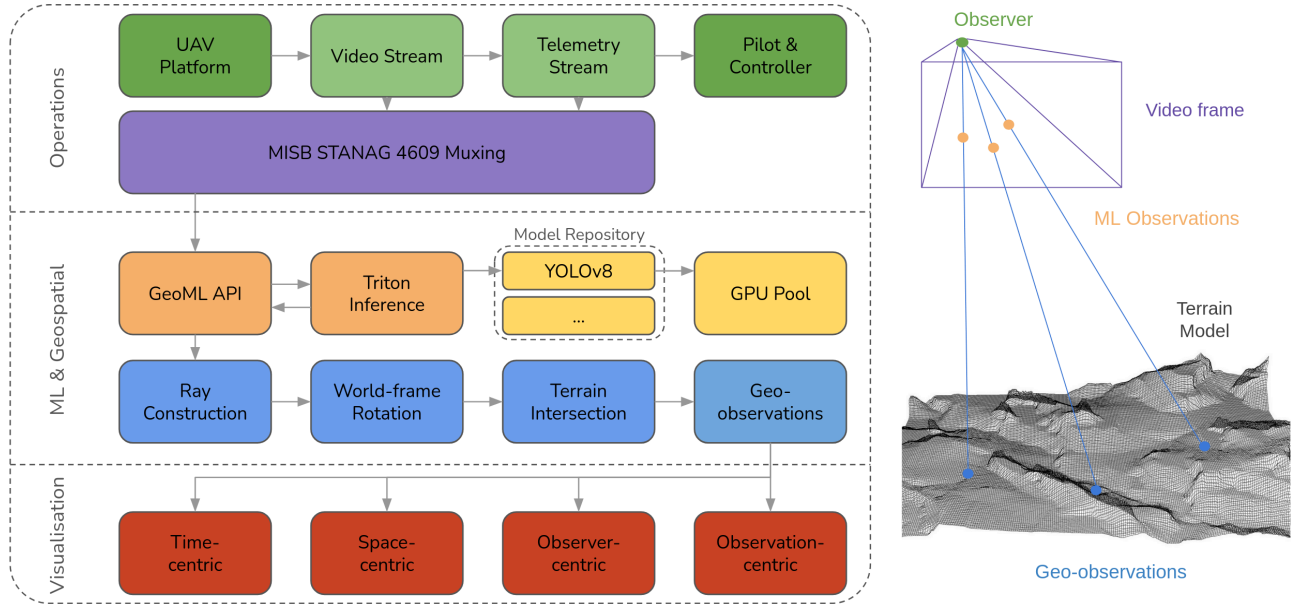


Figure 1: End-to-end GeoML pipeline. Video and telemetry are multiplexed via MISB STANAG 4609 KLV encoding before ingestion by the GeoML FastAPI and NVIDIA Triton for inference. Detections are passed to the geo-positioning pipeline and geo-observations are displayed by four visualisation modes designed with incident response in mind.

3.1 Operations

Tight video-telemetry synchronisation is essential for accurate geolocation. We achieve this by encoding video and telemetry as a single multiplexed stream, following established NATO and defence-imagery standards for full-motion video (FMV), rather than storing them as separate, independently-timestamped files. For this assessment, video and telemetry data were captured using a DJI Matrice 4E with the wide camera lens, then preliminarily multiplexed following NATO FMV Standardisation Agreement (STANAG) 4609, and Motion Imagery Standards Board (MISB) Standard 0601. Video and telemetry synchronisation quality was then as-

essed and updated globally with position and attitude offsets before re-encoding as Key-Length-Value (KLV) metadata within a Video Transport Stream (.ts) format.

The GeoML API is implemented using the FastAPI framework, exposing an interface for inference requests, model management operations, and downstream integration with in-house or third-party visualisation platforms. Inference is managed by an NVIDIA Triton Inference Server, which executes YOLOv8 models concurrently across available GPU resources. Triton’s model repository enables hot-swapping of model versions and independent assignment of model instances to GPUs, enabling multi-mission/-drone deployments. The entire stack is containerised with Docker, enabling horizontal scaling across available resources.

3.2 ML and Geospatial Processing

3.2.1 ML Pipeline

Object detection is performed by YOLOv8-x [Jocher et al., 2023], targeting two operationally relevant classes: *human* and *vehicle*. The extra-large variant was selected to maximise accuracy within the real-time throughput budget afforded by the Triton backend. At operational altitudes, subjects occupy only a small fraction of the image plane, causing standard full-resolution inference to miss detections that fall below the model’s effective receptive field. We apply Slicing Aided Fine-Tuning (SF) and Slicing Aided Hyper Inference (SAHI) [Akyon et al., 2022] to tile each input frame into overlapping sub-windows, run detection independently on each tile, and re-assemble results with class-aware Non-maximum Merging (NMM). Slices are batched before submission to Triton, recovering some GPU utilisation efficiency that would otherwise be lost to the smaller per-tile input size and duplicated processing of overlapping regions.

Both models are trained identically except for the use of sliced data augmentation, isolating the effect of the SF strategy described above; the following details are provided for reproducibility. Two YOLOv8x models are fine-tuned from the same COCO-pretrained checkpoint (Ultralytics v8.4.0) on the VisDrone-DET training set [Zhu et al., 2021]. The baseline model (YOLOv8-x) is trained on full-resolution frames comprising 7,944 training images and 448 validation images, containing 108,794 human and 237,531 vehicle annotations. The SF model’s (YOLOv8-x-SF) training data is augmented with sliced patches: full-resolution training images are partitioned into 640×640 pixel patches with 20% horizontal and vertical overlap, yielding 61,974 training patches and 2,261 validation patches. The increased annotation counts in the slice-augmented dataset (227,238 human; 501,118 vehicle) reflect object duplication across overlapping patch boundaries. Both models are trained for 100 epochs, batch size 32, SGD ($lr = 0.01$, momentum = 0.9). Default Ultralytics YOLOv8 data augmentation is applied throughout, with mosaic, scale, and perspective augmentation disabled.

3.2.2 Geospatial Pipeline

Retrieving the real-world ground location of a detection involves two steps: first, constructing a ray from the drone’s camera through the detected pixel, using the drone’s position and orientation; second, finding where that ray meets the ground, accounting for terrain elevation, Earth’s curvature, and atmospheric refraction. The following paragraphs describe each step in detail.

We first construct a ray from the frame centre, using the UAV’s location and pose together with the sensor’s pose relative to the platform, following the Euler angle order of operations defined in [Motion Imagery Standards Board, 2014]. To account for lens distortion, the camera’s intrinsic parameters were separately estimated using OpenCV and a chessboard calibration pattern [Zhang, 2000]. Using these intrinsics together with the camera’s vertical and horizontal field of view, the 2D pixel coordinates of a detection are converted into an angular offset from the frame centre, giving the direction of the ray corresponding to that detection.

Second, the point at which this ray intersects the ground is then found using a Digital Surface Model (DSM), while also accounting for Earth’s curvature and atmospheric refraction. For simplicity, Earth’s curvature is approximated using the ellipsoidal radius of curvature $R(\varphi, \alpha)$ in a given azimuth α , following Euler’s formula

for normal curvature [Torge and Müller, 2012]:

$$R(\varphi, \alpha) = \frac{M(\varphi) \cdot N(\varphi)}{N(\varphi) \cos^2 \alpha + M(\varphi) \sin^2 \alpha} \quad (1)$$

where $M(\varphi)$ and $N(\varphi)$ are the radius of curvature in the north–south (meridional) and east–west (prime vertical) directions, respectively, at latitude φ .

The impact of atmospheric refraction is approximated with an atmospheric refraction coefficient (k) of 0.13. Combined with the radius of curvature above, the elevation offset (Δh) of the DSM at a given ground distance from the sensor is given by [International Telecommunication Union, Radiocommunication Sector (ITU-R), 2017]:

$$\Delta h(d, \varphi, \alpha) = \frac{d^2}{2R(\varphi, \alpha)}(1 - k) \quad (2)$$

where $R(\varphi, \alpha)$ is the azimuth-dependent radius of curvature from Equation 1, k is the atmospheric refraction coefficient, and d is the ground distance along the ray.

To assess the accuracy of the IFP pipeline, we tracked stationary objects across two flight videos, applying IFP every 10 frames to each tracked object. The resulting position estimate was compared against the object’s true location, identified independently in Esri World Imagery, to compute positioning error. Figure 2 shows the combined results across all 12 tracked objects.

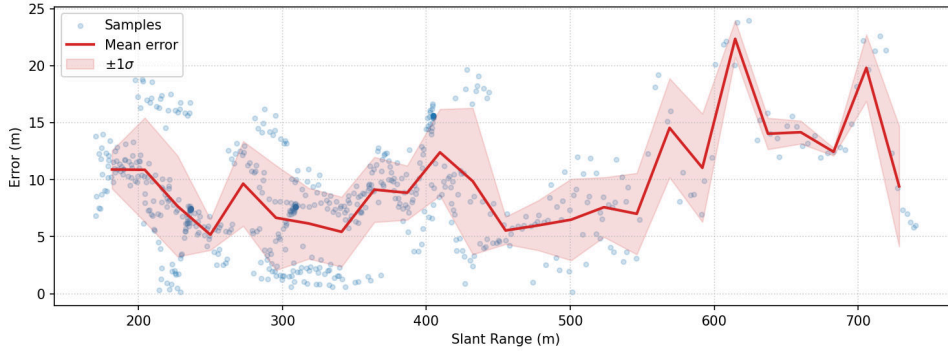


Figure 2: IFP error by distance.

3.3 Visualisation

Even after geo-positioning, the observation stream remains high-dimensional; each detection carries class, confidence, geographic position, drone position, timestamp, and derived attributes. Presenting this raw stream directly to an operator replicates the original information overload problem in a different form. We define four visualisation modes, each reducing the data along a different axis.

The time-centric view (Figure 3a) reduces the video to a one-dimensional activity timeline, allowing operators to identify periods of high activity without reviewing low-content footage, and jump directly to peaks of interest.

The space-centric view (Figure 3b) accumulates all detections across the sequence into a per-class density heatmap overlaid on the base map, making areas of sustained activity immediately legible

The observer-centric view (Figure 3c) displays the drone’s trajectory, colour-coded by detections observed at each position, with the union of all camera footprints outlined to distinguish surveyed from unsurveyed area.

The observation-centric view (Figure 3d) renders each detection as a selectable, colour-coded marker exposing its class, confidence, and geo-location metadata.



Figure 4: Four complementary visualisation modes for drone-based emergency operations.

4 Experiments

To evaluate the performance of the system on both public benchmarks and custom data, we conduct two complementary evaluations. The first is a standard COCO-style benchmark on the VisDrone Det2019 test dataset, providing a reproducible comparison against the broader aerial object detection literature. However, standard COCO size-based stratification into small, medium, and large is of limited value in the aerial domain. In the VisDrone Det2019 test set, 67.7% fall within the small category, 29.1% are medium, and only 3.2% are large. The overwhelming dominance of the small category means that AP_S effectively summarises the vast majority of objects regardless of the underlying operational scenario. Furthermore, COCO size thresholds are not actionable for drone operators: a pilot would have difficulty keeping objects above 32 pixels in width, but can be instructed to fly within 250m. We therefore conduct a second, distance-stratified evaluation on a bespoke held-out dataset with per-detection distance measurements, which directly characterises the operational envelope of each model configuration in terms meaningful for mission planning and pilot guidance.

Both evaluations compare performance across two training strategies - YOLOv8-x fine-tuned on complete full-resolution frames without tiling, and YOLOv8-x-SF fine-tuned on a combination of full-resolution frames and sliced 640×640 patches following the SF strategy recommended by [Akyon et al., 2022]; and three inference modes: Full Inference (FI), in which inference is carried out on unsliced frames; SAHI, in which full frames are partitioned into 640×640 tiles with 20% horizontal and vertical overlap and a class-aware NMM step resolves overlapping detections across tile boundaries; and FI + SAHI, in which sliced predictions are first merged via NMM before being combined with full-frame predictions in a second NMM pass. For both experiments, metrics are computed at an IoU threshold of 0.5. FPS is reported as inference throughput, measured on an NVIDIA RTX 6000 Ada Generation GPU.

4.1 VisDrone Det2019 Benchmark

Both models are evaluated on the VisDrone Det2019 test dataset under all three inference modes, following the standard COCO evaluation protocol. VisDrone annotations are remapped to two classes: human (VisDrone categories 1–2) and vehicle (categories 3–10); ignored regions (score= 0) and the others category (11) are excluded. This provides a standardised comparison of the effects of training strategy and inference mode on a widely used aerial detection benchmark.

4.2 Distance-Stratified Evaluation

The distance-stratified evaluation dataset was collected from two flight sequences over a rural area in Ireland using a DJI Matrice 4E, capturing video at 1920×1080 resolution and 30 FPS. Frames were extracted at 30-frame intervals, yielding 213 images across 3988 annotated object instances. Ground-truth bounding boxes were annotated manually, with slant range, i.e. direct diagonal distance not ground distance to the target, computed per annotation by projecting the bounding box centres using in-frame positioning as described in Section 3. The dataset distribution is shown in the rightmost column of Figure 5.

For each image, predictions are matched to ground-truth annotations using greedy IoU matching at a threshold of 0.5, with predictions ranked by confidence score. Precision, recall, and AP are computed per image per band and averaged across images. We report both discrete 100 m bands and cumulative bands of the form $[0, d_{\max})$, which show how aggregate performance changes as the distance ceiling is raised. Predictions from the full image are used as candidates regardless of their own projected distance, since the band assignment is defined by the ground-truth object rather than filtering predictions by their own estimated range, treating distance as a known measurement.

5 Results

5.1 VisDrone Det2019 Benchmark

Model	AP ₅₀	AP _S	AP _M	AP _L	AR _S	AR _M	AR _L	FPS
YOLOv8-x + FI	0.459	0.172	0.482	0.626	0.209	0.544	0.685	90.51
YOLOv8-x + SAHI	0.447	0.190	0.365	0.263	0.250	0.423	0.304	24.53
YOLOv8-x + FI + SAHI	0.479	0.178	0.428	0.529	0.238	0.509	0.634	18.76
YOLOv8-x-SF + FI	0.456	0.160	0.510	0.729	0.194	0.571	0.792	86.33
YOLOv8-x-SF + SAHI	0.587	0.256	0.531	0.609	0.312	0.598	0.691	20.37
YOLOv8-x-SF + FI + SAHI	0.551	0.213	0.509	0.621	0.270	0.582	0.717	15.03

Table 1: Mean COCO metrics averaged across all distance bands and classes, with inference throughput (FPS) measured on NVIDIA RTX 6000 Ada Generation.

Table 1 reports COCO detection metrics on the VisDrone Det2019 test set. YOLOv8-x-SF + SAHI achieves the strongest overall performance, attaining the highest AP (0.587), AP_S (0.256), AP_M (0.531), AR_S (0.312), and AR_M (0.598), demonstrating that sliced fine-tuning combined with sliced inference is the most effective configuration for this domain.

Full-frame inference consistently yields higher AP_L and AR_L, which is expected given that sliced inference is not optimised for large objects. The combined FI + SAHI mode improves over full-frame inference alone on AP but falls short of SAHI-only across most metrics.

Throughput varies substantially across inference modes: full-frame inference achieves approximately 86 FPS, while SAHI reduces this to around 20 FPS due to the overhead of slicing and NMM post-processing.

5.2 Distance-Stratified Evaluation

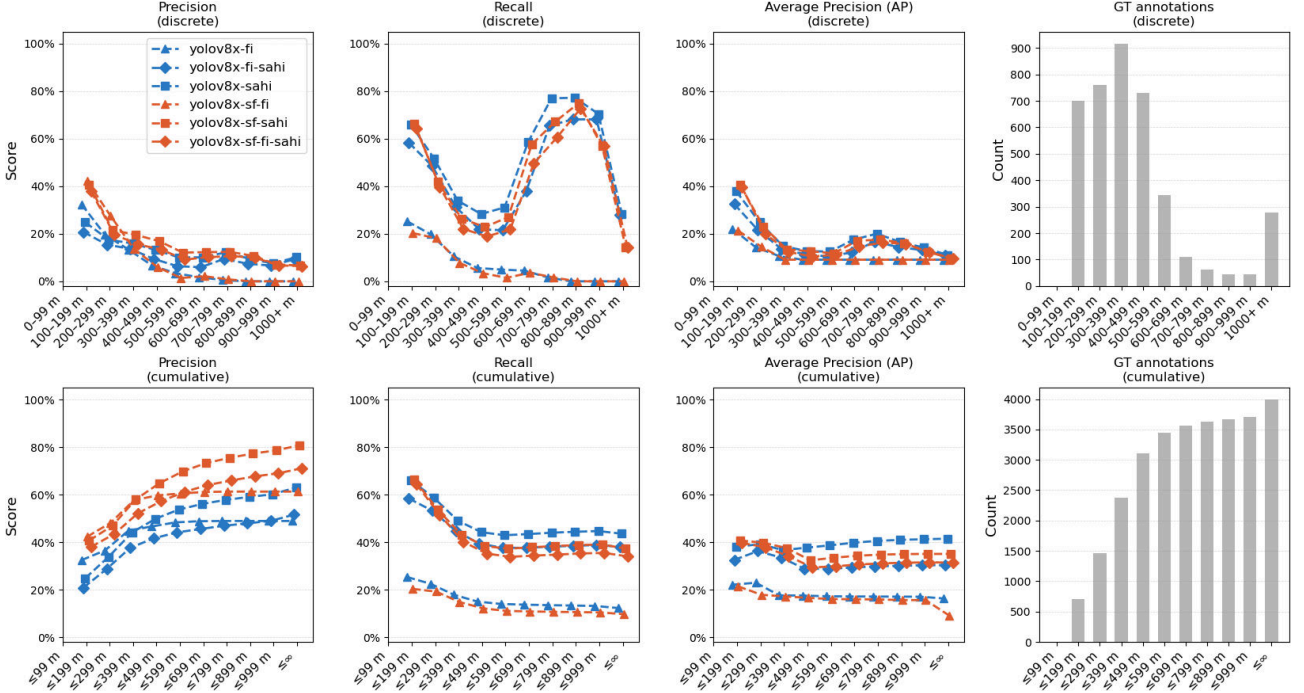


Figure 5: Mean detection performance per distance band at IoU=0.5, averaged across Human and Vehicle classes.

Figure 5 shows precision, recall, and AP stratified by slant range in 100 m discrete bands and cumulative bands up to 1 km.

Because ground-truth slant range is itself derived from the IFP pipeline rather than measured independently, the distance axis inherits IFP positioning error (up to approximately 25 m, Figure 2) rather than being an exact reference. Relative to the 100 m discrete bin width used in Figure 5, this error is large enough to shift the band assignment of detections near a bin boundary, but by no more than one band, while detections near bin centres are effectively unaffected. We therefore expect the reported per-band metrics to be subject to some boundary blur rather than systematic bias, and interpret the cumulative bands in Figure 5, which are less sensitive to single-band reassignment, as the more robust indicator of the operational envelope.

Across both model families, SAHI inference consistently improves recall beyond approximately 200 m, consistent with [Akyon et al., 2022]: at ranges where vehicles subtend only a small fraction of the full frame, localised slice-level context improves sensitivity relative to full-frame inference. This recall gain comes at the cost of precision on near-field objects, the expected fragmentation effect whereby large nearby objects span multiple slices, producing overlapping predictions not fully suppressed by NMM postprocessing.

The combined inference mode FI + SAHI, which applies NMM first to slice predictions and then merges with full-frame predictions in a second pass, does not substantially improve near-field precision over SAHI alone, suggesting the fragmentation effect persists across both NMM stages. The discrete band plots show a recall dip at medium range (300–500 m) and recovery at longer distances, which may reflect performance being dominated by a small number of salient objects at long range.

We note that both the distance-stratified evaluation and the IFP error validation underpinning it are based on limited data: 213 images from two flight sequences at a single site, and IFP error estimated from only 12 tracked objects. The trends reported above should therefore be considered preliminary; a larger and more varied evaluation dataset would be needed to establish firm conclusions.

Peak RAM usage is uniform across all three inference modes at ~1600 MB, whereas sliced inference showed an increase in peak VRAM usage (~2260 MB vs. ~450 MB at batch size 16).

6 Conclusion

We have presented a framework that addresses the situational awareness bottleneck in drone-supported emergency response. The VisDrone benchmark confirms the benefits of slicing at both training and inference time, with YOLOv8-x-SF + SAHI the most effective configuration. Our preliminary distance-based evaluation suggests a detection range of approximately 500 m and shows that SAHI consistently improves recall and precision as distance increases, offering an initial characterisation of the operational envelope for mission planning.

The four visualisation modes offer a principled decomposition of the high-dimensional observation stream, each targeting a distinct operator query pattern: *when was activity observed?*, *where is it concentrated?*, *what area did the drone cover?* and *what is this object doing?*

Future work will prioritise dataset quality and coverage as a prerequisite for validating the operational envelope with respect to extended conditions such as smoke/haze, low-light, and differing UAV and sensor platforms. In the longer term, geo-coded annotations will address the gap between existing datasets and real deployments, exploring combined visual and spatial tracking and re-identification, and behaviour and anomaly detection to extract higher-order intelligence from data.

Acknowledgments

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